

Total No. of Questions : 8]

SEAT No. :

PC2454

[Total No. of Pages : 2

[6354]-582

B.E. (Information Technology)

WIRELESS COMMUNICATION

(2019 Pattern) (Semester - VII) (Elective - IV) (414445 D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) How Code Division Multiple Access (CDMA) works? Explain in detail with suitable diagram. **[9]**

b) What do you understand by OFDM (Orthogonal frequency Division Multiplexing)? Elaborate in detail with example. **[9]**

OR

Q2) a) How Time Division Multiple Access (TDMA) is useful in wireless communication? Justify **[9]**

b) What is MIMO? Explain two formats of MIMO. **[9]**

Q3) a) Draw and explain WAP Architecture in detail. **[9]**

b) Explain the terms: Wi-Fi Direct, Li-Fi, NFC, Sigfox. **[8]**

OR

Q4) a) Explain the wireless Application Protocol and WAP Programming Model in detail. **[9]**

b) Explain the terms : Z- Wave, LoRaWAN, RT Wi-Fi, SPEED. **[8]**

P.T.O.

- Q5) a)** Discuss UMTS Security and Bluetooth Security in detail. **[9]**
- b)** What security issues and challenges are there in 1G, 2G, 3G, and 4G communication model? **[9]**

OR

- Q6) a)** Discuss Multimedia security in 5G and 6G in detail. **[9]**
- b)** What do you know about kismet, URH (Universal Radio Hacker)? Give detail. **[9]**

- Q7) a)** Explain 5G NR working and give its benefits. **[9]**
- b)** How Simultaneous Transmission and Reflection (STAR) for 360° Coverage is used in communication. **[8]**

OR

- Q8) a)** What are the Applications of Wireless Technology? Enlist and explain in brief. **[9]**
- b)** What do you understand by Holographic MIMO Surfaces for 6G Wireless Networks? **[8]**

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